

# SOURDOUGH BREAD

YIELD: 2 loaves (~900 g each)

February 23, 2026 · Version 1 · Side Dish

Lean naturally leavened bread with staged fermentation, folds, and covered baking.

## PREPARATION

Scaling basis: Bread flour · 1100 g

| INGREDIENT | QUANTITY | SCALING | PROCEDURE |
|------------|----------|---------|-----------|
|------------|----------|---------|-----------|

### PHASE A — BUILD LEVAIN

|                                                          |       |       |                                                     |
|----------------------------------------------------------|-------|-------|-----------------------------------------------------|
| <u>Mature sourdough starter</u><br>Inoculate the levain. | 20 g  | 1.82% | 1. Mix starter, flour, and water until homogeneous. |
| <u>Bread flour</u><br>Feed the levain.                   | 100 g | 9.09% | 2. Cover and ferment until expanded and active.     |
| <u>Water</u><br>Hydrate the levain.                      | 100 g | 9.09% |                                                     |

### TECHNIQUE APPLICATIONS

| Technique           | Controls                                                         | Purpose              |
|---------------------|------------------------------------------------------------------|----------------------|
| <u>Levain build</u> | Ferment about 8 h at room temperature until expanded and active. | Produce ripe leaven. |

### PRINCIPLES

- Yeast and lactic acid bacteria fermentation — The culture produces gas and acidity.

### PHASE OUTPUTS

| Key         | Phase Output | Description                  |
|-------------|--------------|------------------------------|
| ripe-levain | Ripe levain  | Active leaven for the dough. |

PHASE B — MIX AND DEVELOP DOUGH

|                                                                                   |        |        |                                                                                                           |
|-----------------------------------------------------------------------------------|--------|--------|-----------------------------------------------------------------------------------------------------------|
| <u>Bread flour</u><br>Mix into the dough.                                         | 1000 g | 90.91% | 3. Combine water, ripe levain, and flour; cover and rest.                                                 |
| <u>Water</u><br>Use 600 g to hydrate the flour and reserve 40 g to dissolve salt. | 640 g  | 58.18% | 4. Dissolve salt in the additional water, mix it into the dough, and perform three stretch-and-fold sets. |
| <u>Salt</u><br>Dissolve in the additional water.                                  | 20 g   | 1.82%  | 5. Let the dough bulk ferment until visibly aerated.                                                      |

TECHNIQUE APPLICATIONS

| Technique                | Controls                                     | Purpose                                      |
|--------------------------|----------------------------------------------|----------------------------------------------|
| <u>Autolyse</u>          | Rest flour and water 30 min to 4 h.          | Hydrate flour before salt incorporation.     |
| <u>Folding</u>           | Perform 3 stretch-and-fold sets over 90 min. | Strengthen dough without intensive kneading. |
| <u>Bulk fermentation</u> | Ferment 2–3 h until visibly aerated.         | Develop gas and dough structure.             |

PRINCIPLES

- Hydration ratio — Water proportion controls dough handling and crumb.
- Gluten development — Resting and folding develop structure.
- Salt control of fermentation — Salt affects fermentation timing.

PHASE OUTPUTS USED

| Phase Output | Use                                                   |
|--------------|-------------------------------------------------------|
| Ripe levain  | Dissolve <span>ripe-levain</span> in the dough water. |

PHASE OUTPUTS

| Key        | Phase Output         | Description                    |
|------------|----------------------|--------------------------------|
| bulk-dough | Bulk-fermented dough | Aerated dough ready to divide. |
|            |                      |                                |

## PHASE C — SHAPE, PROOF, AND BAKE

6. Divide the dough into two, pre-shape, rest 30 min, shape, and proof 2–3 h at room temperature or overnight in the refrigerator.
7. Preheat a lidded casserole at 240 C for at least 20 min.
8. Bake each scored loaf covered, uncover for the final 10 min, and cool completely before slicing.

### TECHNIQUE APPLICATIONS

| Technique            | Controls                                                                    | Purpose                                 |
|----------------------|-----------------------------------------------------------------------------|-----------------------------------------|
| <u>Staged baking</u> | Bake covered then uncover for the last 10 min; total about 40 min at 240 C. | Allow spring before browning the crust. |

### PRINCIPLES

- Steam-assisted oven spring — Covered baking delays crust setting during expansion.
- Maillard reaction — The uncovered finish browns the crust.

### PHASE OUTPUTS USED

| Phase Output         | Use                                                                  |
|----------------------|----------------------------------------------------------------------|
| Bulk-fermented dough | Divide <b>bulk-dough</b> , pre-shape, rest, shape, proof, and score. |

### FAILURE MODES

| Symptom     | Likely cause                                  | Corrective action                                |
|-------------|-----------------------------------------------|--------------------------------------------------|
| Dense loaf  | Levain or bulk fermentation was insufficient. | Extend fermentation and confirm levain activity. |
| Flat dough  | Gluten was weak or dough overproofed.         | Add a fold set and shorten final proof.          |
| Gummy crumb | Centre was underbaked or sliced hot.          | Bake to about 98 C and cool fully.               |